Jobs and DSA

Microsoft SWE Intern
https://apply.careers.microsoft.com/careers/job/1970393556625300

Amazon Software Engineer
Software Dev Engineer I, Amazon University Talent Acquisition - Job ID: 2935068

Google Software Engineer

https://www.metacareers.com/profile/job_details/1503103504068487/

Intuit
https://jobs.intuit.com/job/-/-/27595/88105818816

Stripe SWE Intern

https://stripe.com/jobs/listing/software-engineer-intern/6230907

1. Two Pointers

Focus: Reducing nested loops by moving pointers from different ends or different speeds.

#	Problem	Difficulty
1	Two Sum II - Input Array is Sorted	Medium

2	Container With Most Water	Medium
3	Trapping Rain Water	Hard
4	3Sum	Medium
5	Squares of a Sorted Array	Easy
6	Valid Palindrome	Easy

2. Sliding Window

Focus: Subarrays or substrings of specific size or constraints.

#	Problem	Difficulty
7	Maximum Sum Subarray of Size K	Easy
8	Longest Substring Without Repeating Characters	Medium
9	Minimum Window Substring	Hard
10	Longest Repeating Character Replacement	Medium
11	Permutation in String	Medium

12	Fruits Into Baskets	Medium

3. Cyclic Sort

Focus: Arrays containing numbers in a given range (1 to n).

#	Problem	Difficulty
13	Missing Number	Easy
14	Find All Numbers Disappeared in an Array	Easy
15	Find the Duplicate Number	Medium
16	First Missing Positive	Hard
17	Set Mismatch	Easy

4. Reversal of a Linked List (In-place)

Focus: Manipulating pointers to reverse order without extra memory.

#	Problem	Difficulty
18	Reverse Linked List	Easy

19	Reverse Linked List II	Medium
20	Reverse Nodes in k-Group	Hard

5. Stack (Monotonic & Traversal)

Focus: Finding next greater/smaller elements or parsing expressions.

#	Problem	Difficulty
21	Valid Parentheses	Easy
22	Daily Temperatures	Medium
23	Next Greater Element II	Medium
24	Largest Rectangle in Histogram	Hard
25	Min Stack	Medium
26	Asteroid Collision	Medium

6. Hash Maps

Focus: Frequency counting, index mapping, and caching.

#	Problem	Difficulty
27	Two Sum	Easy
28	Group Anagrams	Medium
29	Longest Consecutive Sequence	Medium
30	Subarray Sum Equals K (Prefix Sum + Hash Map)	Medium
31	Ransom Note	Easy
32	Design HashMap	Easy

7. BFS and DFS (Graph Traversal)

Focus: Visiting nodes in a graph, shortest paths (unweighted), or exhaustive search.

#	Problem	Difficulty
33	Clone Graph	Medium
34	Word Ladder	Hard
35	01 Matrix	Medium

36	Rotten Oranges	Medium
37	Number of Connected Components in an Undirected Graph	Medium
38	Graph Valid Tree	Medium

8. Tree Traversals

Focus: Pre/In/Post-order, recursion, and stack-based iteration.

#	Problem	Difficulty
39	Binary Tree Level Order Traversal	Medium
40	Maximum Depth of Binary Tree	Easy
41	Diameter of Binary Tree	Easy
42	Lowest Common Ancestor of a Binary Tree	Medium
43	Binary Tree Maximum Path Sum	Hard
44	Validate Binary Search Tree	Medium
45	Serialize and Deserialize Binary Tree	Hard

46	Path Sum II	Medium
47	Kth Smallest Element in a BST	Medium

9. Island Finding (Grid Traversal)

Focus: Matrix traversal to find connected components (Implicit Graph).

#	Problem	Difficulty
48	Number of Islands	Medium
49	Max Area of Island	Medium
50	Number of Closed Islands	Medium
51	Surrounded Regions	Medium
52	Island Perimeter	Easy

10. Two Heaps

Focus: Managing data streams to find median or partitioning data.

#	Problem	Difficulty

53	Find Median from Data Stream	Hard
54	Sliding Window Median	Hard
55	IPO	Hard
56	Find Right Interval	Medium

11. Subsets (Backtracking)

Focus: Power sets and permutations.

#	Problem	Difficulty
57	Subsets	Medium
58	Subsets II (With Duplicates)	Medium
59	Permutations	Medium
60	Letter Case Permutation	Medium
61	Generalized Abbreviation	Medium

12. Binary Search on Answer

Focus: Searching within a range of possible answers rather than an array.

#	Problem	Difficulty
62	Koko Eating Bananas	Medium
63	Capacity To Ship Packages Within D Days	Medium
64	Split Array Largest Sum	Hard
65	Minimum Number of Days to Make m Bouquets	Medium
66	Magnetic Force Between Two Balls	Medium

13. Binary Search on Array

Focus: Standard and modified binary search on sorted/rotated arrays.

#	Problem	Difficulty
67	Binary Search	Easy
68	Search in Rotated Sorted Array	Medium
69	Find Minimum in Rotated Sorted Array	Medium
70	Find Peak Element	Medium

71	Find First and Last Position of Element in Sorted Array	Medium
72	Search Insert Position	Easy

14. Trie (Prefix Tree)

Focus: String retrieval, autocomplete, and prefix matching.

#	Problem	Difficulty
73	Implement Trie (Prefix Tree)	Medium
74	Design Add and Search Words Data Structure	Medium
75	Word Search II	Hard
76	Replace Words	Medium
77	Longest Word in Dictionary	Medium

15. Disjoint Set Union (Union-Find)

Focus: Connectivity, merging sets, and cycle detection.

#	Problem	Difficulty

78	Number of Provinces	Medium
79	Redundant Connection	Medium
80	Accounts Merge	Medium
81	Number of Operations to Make Network Connected	Medium
82	Smallest String With Swaps	Medium

16. Top K Elements (Heap)

Focus: finding the k-largest/smallest/frequent items.

#	Problem	Difficulty
83	Kth Largest Element in an Array	Medium
84	Top K Frequent Elements	Medium
85	K Closest Points to Origin	Medium
86	Sort Characters By Frequency	Medium
87	Reorganize String	Medium

17. Backtracking (General Constraints)

Focus: Exhaustive search with pruning (excluding standard subset problems).

#	Problem	Difficulty
88	Combination Sum	Medium
89	Combination Sum II	Medium
90	Palindrome Partitioning	Medium
91	Word Search	Medium
92	N-Queens	Hard
93	Sudoku Solver	Hard

18. Dynamic Programming

Focus: Breaking problems into subproblems (1D, 2D, Knapsack, Strings).

#	Problem	Difficulty
94	Climbing Stairs	Easy
95	House Robber	Medium

96	Coin Change	Medium
97	Longest Increasing Subsequence	Medium
98	Longest Common Subsequence	Medium
99	Word Break	Medium
100	Partition Equal Subset Sum	Medium
101	Unique Paths	Medium
102	Edit Distance	Hard
103	Best Time to Buy and Sell Stock	Easy
104	Maximum Product Subarray	Medium
105	Burst Balloons	Hard

19. Topological Sort

Focus: Linear ordering of nodes (dependencies, scheduling).

#	Problem	Difficulty

106	Course Schedule	Medium
107	Course Schedule II	Medium
108	Alien Dictionary	Hard
109	Sequence Reconstruction	Medium
110	Parallel Courses	Medium

20. Bitwise XOR Tricks

Focus: Manipulating bits to find unique elements or save space.

#	Problem	Difficulty
111	Single Number	Easy
112	Single Number II	Medium
113	Single Number III	Medium
114	Counting Bits	Easy
115	Sum of Two Integers	Medium